CYBERSECURITY LOG ANALYSIS REPORT

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Activity: CTM-ACTIVITY-8

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ATTACK 1 ANALYSIS

1. Identified Attack Type: SSH Brute-Force Attack

Attack analysis: The logs show repeated SSH authentication attempts against the same destination server from one source IP. The attacker tried several account names in a short period, which is consistent with password guessing or credential guessing.

2. Three Supporting Log Evidence Entries

Evidence 1: Event ID / Timestamp: BF-0001 — 2026-10-31 08:14:02 UTC

Observed activity: Source 198.51.100.23 attempted to log in to 192.0.2.10 on TCP port 22 using the admin account.
The login failed.

Why it matters: This is the first recorded SSH authentication failure from the same source.

Evidence 2: Event ID / Timestamp: BF-0003 — 2026-10-31 08:14:12 UTC

Observed activity: The same source tried the administrator username. The event was marked high severity and noted multiple username attempts.

Why it matters: Trying different privileged account names within seconds suggests systematic account guessing rather than an ordinary isolated mistake.

Evidence 3: Event ID / Timestamp: BF-0005 and BF-0007 — 2026-10-31 08:14:22 and 08:16:30 UTC

Observed activity: After five attempts, the admin account was temporarily locked. Later, the source was rate-limited after the sixth attempt.

Why it matters: The security controls blocked further attempts, showing that protective responses were triggered.

3. Attack Analysis

The logs show a series of SSH login failures coming from one source IP address, 198.51.100.23, and targeting the same server. The source tried the usernames admin, administrator, and root within a short period. This repeated pattern is consistent with a brute-force or password-guessing attempt. The account lockout and rate limit also show that the system responded to the suspicious activity. There is a separate successful login in BF-0008, but it came from 203.0.113.44 using the student account, not from the suspected attacker. It should be checked for authorization, but it does not prove that the brute-force source got in.

4. Attack Status

Classification: UNSUCCESSFUL.

I classified this attack as unsuccessful because the repeated attempts from 198.51.100.23 were recorded as failed or blocked. BF-0005 shows the account lockout, and BF-0007 shows that the source was rate-limited. No successful login from the suspected attacker appears in the provided records.

5. Possible Scenario if the Attack Succeeds

If the attacker eventually guessed a valid password, they could gain remote access to the server. They might view or copy private files, change settings, create unauthorized accounts, install harmful software, or misuse the server to attack other systems. If the compromised account had administrator privileges, the attacker could gain much greater control over the machine.

6. Prevention and Mitigation

Recommendation 1: Require multi-factor authentication for remote access and, where possible, disable password-based SSH login in favor of secure SSH keys.

Recommendation 2: Use strong and unique passwords. Keep account lockout and rate-limiting controls enabled so repeated login attempts are slowed or blocked.

Recommendation 3: Limit SSH access to trusted IP addresses or a secure VPN. Disable direct root login and regularly review authentication logs for repeated failures and unfamiliar successful logins.

ATTACK 2 ANALYSIS

1. Identification of Attack: Phishing Attack (credential-harvesting attempt)

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: PH-0001 — 2026-10-31 09:02:11 UTC

Observed activity: An email titled “Urgent: Password Expiration” was delivered to employee01@example.test. It warned that the account would be disabled unless the employee verified their credentials.

Why it matters: The urgent warning pressures the employee to act quickly. This is a common social-engineering technique used in phishing messages.

Evidence 2: Event ID / Timestamp: PH-0003 — 2026-10-31 09:05:18 UTC

Observed activity: The employee clicked the link to account-verify.example.test. The event was recorded as link_clicked and marked high severity.

Why it matters: This confirms that the recipient interacted with the suspicious email. It means the phishing attempt reached the user, although clicking alone does not prove that credentials were stolen.

Evidence 3: Event ID / Timestamp: PH-0004 and PH-0005 — 2026-10-31 09:05:41 and 09:06:10 UTC

Observed activity: The browser detected a page that resembled a login form and blocked access. The security gateway later quarantined the email.

Why it matters: The fake login page suggests an attempt to collect credentials. The block and quarantine show that security controls stopped the user from proceeding to that page.

3. Attack Analysis

The first email used a false password-expiration warning and a link whose domain did not match the organization's official domain. The employee clicked the link, so the message was able to influence the recipient. However, the browser blocked the page when it detected a login form, and the email security gateway quarantined the message. The dataset also shows a separate unexpected HTML attachment sent to employee02; that attachment was blocked by the email security policy. These events support a

phishing attempt, but the logs do not show that any employee entered a password or that an account was taken over.

4. Attack Status

Classification: PARTIALLY SUCCESSFUL.

The phishing attempt was partially successful because the email was delivered and employee01 clicked its suspicious link. However, the fake login page was blocked and the email was quarantined. There is no recorded event showing that the employee submitted credentials or that an account was compromised. The attacker reached the user, but the available logs do not show successful credential theft.

5. Possible Scenario if the Attack Succeeds

If the employee had entered their username and password on the fake page, the attacker could use those details to access the employee's email or other work accounts. The attacker might read confidential messages, steal company files, impersonate the employee, change payment information, or send more phishing emails to other staff members. Reused passwords could also put additional accounts at risk.

6. Prevention and Mitigation

Recommendation 1: Train employees to recognize suspicious messages, especially those that create urgency or ask for passwords. Encourage them to check the sender and website address and to report questionable emails.

Recommendation 2: Use email filtering and browser protection that can detect suspicious links and attachments. Keep these tools updated and quarantine messages that match phishing indicators.

Recommendation 3: Enable multi-factor authentication and require unique passwords. If an employee may have entered credentials, reset the password, revoke active sessions, and review the account's sign-in history immediately.

ATTACK 3 ANALYSIS

1. Identification of Attack: Distributed Denial-of-Service (DDoS) attack pattern

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: DOS-0002 — 2026-10-31 10:20:10 UTC

Observed activity: The server received 420 requests per second from 198.51.100.90, while the normal baseline was 35 requests per second. Rate limiting was activated and the server returned HTTP 429.

Why it matters: The traffic was twelve times the normal baseline. This unusual increase is a warning sign that the service may be receiving excessive requests.

Evidence 2: Event ID / Timestamp: DOS-0004 — 2026-10-31 10:20:30 UTC

Observed activity: Traffic reached 850 requests per second from 198.51.100.92. The log recorded increased service latency and HTTP 503 responses.

Why it matters: The high traffic was affecting service performance. HTTP 503 means the service was temporarily unable to handle some requests.

Evidence 3: Event ID / Timestamp: DOS-0005 and DOS-0006 — 2026-10-31 10:20:40 and 10:20:50 UTC

Observed activity: The health check failed while traffic remained at 850 requests per second. The system then enabled traffic filtering and rate limiting.

Why it matters: The failed health check confirms an availability problem. The mitigation entry shows that defensive measures were applied to reduce the impact.

3. Attack Analysis

The logs show a sharp rise in requests to the same HTTPS service at 192.0.2.20 on port 443. The request rate increased from a normal 35 requests per second to 850 requests per second, and the traffic came from several source IP addresses. The server began returning HTTP 429 and 503 responses, and one health check failed. These patterns are consistent with a DDoS-style traffic flood intended to make a service difficult to access. After filtering and rate limiting were enabled, traffic decreased to 160 requests per second and then to 42 requests per second. The final health check returned HTTP 200, showing that the service recovered.

4. Attack Status

Classification: PARTIALLY SUCCESSFUL.

I classified the attack as partially successful because it caused a temporary availability problem. The HTTP 503 responses and failed health check show that the service was affected. However, the mitigation controls reduced the traffic, and DOS-0008 records a successful health check with HTTP 200. The logs therefore show a temporary disruption followed by recovery, rather than a continuing outage.

5. Possible Scenario if the Attack Succeeds

If the attack continued and overwhelmed the server or network capacity, legitimate users might not be able to open the website or use its online services. The organization could lose sales or work time, delay transactions, disappoint customers, and damage its reputation. A prolonged outage could also affect other services that depend on the targeted system.

6. Prevention and Mitigation

Recommendation 1: Use a DDoS protection service, such as a reputable content delivery network or cloud-based traffic scrubbing service, to filter malicious traffic before it reaches the server.

Recommendation 2: Configure rate limits, firewall rules, connection limits, and traffic monitoring. Set alerts for unusual request rates, rising latency, and repeated HTTP 5xx errors.

Recommendation 3: Prepare an incident-response and recovery plan. Monitor server capacity and service health, and regularly test whether traffic filtering and backup capacity can keep essential services available.

FINAL REFLECTION

1. Among the three attacks, which could cause the greatest damage to an organization? Explain your answer.

For me, a successful phishing attack could cause the greatest long-term damage if it allows an attacker to steal an employee's account. With access to a real account, the attacker may read confidential information, impersonate the employee, and move

further into the organization's systems. A DDoS attack can cause serious disruption, especially while it is happening, but stolen credentials may give an attacker access that lasts longer if it is not detected. The actual damage still depends on which system is targeted and how quickly the organization responds.

2. Why is analyzing system logs important in detecting and responding to cybersecurity incidents?

Analyzing system logs is important because logs provide a record of activities happening in a system. They help analysts see when an event occurred, where it came from, what account or service was involved, and whether the action was allowed or blocked. By checking these details, analysts can recognize unusual patterns, understand what happened, and respond before the problem becomes worse. Logs also help prevent us from making conclusions without enough evidence.

3. What is the most important cybersecurity lesson you learned from this activity?

The most important lesson I learned is that not every suspicious activity means that a system has already been compromised. For example, a failed login is not the same as a successful break-in, and clicking a phishing link does not automatically mean that a password was stolen. We need to examine the complete log evidence, check the result of each event, and understand whether the security controls worked. As a future IT professional, I should investigate carefully before making a conclusion.

SUBMISSION CHECKLIST

☑ I analyzed all three CSV datasets.

☑ I identified each attack using evidence from the logs.

☑ I included at least three supporting log observations for each attack.

☑ I explained the attack status using specific log entries.

☑ I described a possible scenario if each attack succeeds.

☑ I provided at least three relevant security recommendations for each attack.

☑ I completed all three final reflection questions.